

rf-frontend: Design, Verification and Release Evidence for Hardware Revision A

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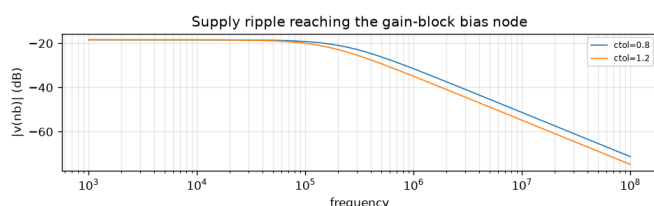


Fig. 1. Circuit simulation at declared corners: P-1

Abstract—This paper documents the requirements, design method, verification and release evidence of a pcb-scope electronic product. The design was developed with an iterative modify–verify–retain loop (11 retained of 13 candidate changes) and gated by electrical-rule, design-rule, corner simulation and rule-based checks. The current gate result is G3 BLOCKED.

Index Terms—hardware design, verification, PCB, design rules, evidence

I. INTRODUCTION

State the problem, intended use and the contribution. Sources consulted during design: [1] [2] [3] [4] [5] [6] [7] [8] [9].

II. REQUIREMENTS

III. DESIGN METHOD

Describe the architecture and the design rules applied (see Table II).

IV. VERIFICATION

Every check executed is recorded as a hashed receipt (Table III); the closest call of each, relative to its limit, is in Table IV.

V. ITERATION RESULTS

VI. CONCLUSION

Summarize what was demonstrated, the remaining blockers and the intended next gate.

REFERENCES

- [1] B. R. Archambeault, *PCB Design for Real-World EMI Control*. Kluwer Academic / Springer, 2002.
- [2] E. Bogatin, *Signal and Power Integrity—Simplified*, 3rd ed. Prentice Hall/Pearson, 2018.
- [3] C. Bowick, J. Blyler, and C. Ajluni, *RF Circuit Design*, 2nd ed. Newnes/Elsevier, 2008.
- [4] D. Brooks and J. Adam, *PCB Design Guide to Via and Trace Currents and Temperatures*. Artech House, 2021.



Fig. 2. Assertion margins of every check: margins

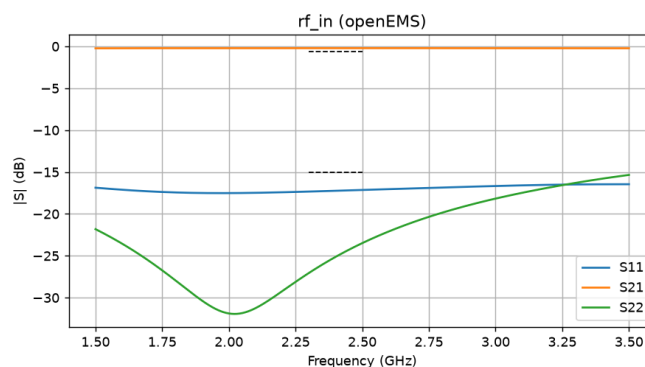


Fig. 3. openEMS S-parameters of the board copper: rf_in

- [5] P. Horowitz and W. Hill, *The Art of Electronics*, 3rd ed. Cambridge

TABLE I
REQUIREMENTS BASELINE

id	statement	units	conditions	method	gate
HR-001	RF input and output networks: return loss $\geq$ 15 dB and insertion loss $\leq$ 0.6 dB from 2.3 to 2.5 GHz (connector landing to gain-block pin)	dB	full-wave model of the released copper, 50 ohm ports	simulation	G3
HR-002	Every RF line 50 ohm +/-10 % on the fabricator stackup	ohm	field solution of the routed cross-section incl. mask and coplanar gaps	analysis	G3
HR-003	Gain-block bias current 100 mA +/-10 % from the 12 V input (device voltage 4.4 V typ, datasheet value to confirm)	mA	12 V +/-10 % input, 25 C	analysis	G3
HR-004	Every resistor at or below 50 % of its rated power (2x derating)	%	worst-case 13.2 V input, 70 C ambient rating	analysis	G3
HR-005	25 MHz clock into a 50 ohm load: overshoot $\leq$ 15 %, ringback $\leq$ 20 %, settled within 5 ns	%	routed clock net, 1 ns edges	simulation	G3
HR-006	+3V3 rail impedance at the oscillator below Z_target (2 % ripple on a 20 mA step) from 1 kHz to 100 MHz	ohm	released capacitor set and planes	analysis	G3
HR-007	Every dissipating part at least 20 C below its maximum junction/element temperature	C	40 C still air, board horizontal	analysis	G3
HR-008	Clock-harmonic radiated-emission estimate at least 6 dB below CISPR 32 Class B (estimate only; chamber test at EVT)	dB	routed clock loops, 10 m limit line	analysis	G3
HR-009	12 V input path: trace temperature rise $\leq$ 10 C at 150 mA	C	IPC-2152 external/internal fits	analysis	G3
HR-010	Schematic connectivity equals the golden circuit description	nets	fresh KiCad netlist export	inspection	G3
HR-011	Assembled board: measured S11 $\leq$ -10 dB from 2.3 to 2.5 GHz at both RF ports	dB	VNA, calibrated at the SMA reference planes	test	G4

TABLE II
DESIGN DECISIONS AND THE RULES THAT JUSTIFY THEM

id	topic	decision	rules	sources
D-1	RF line	0.32 mm grounded coplanar, 0.3 mm pour gap, 0.2 mm prepreg to In1 GND; 49.7 ohm	-	tools/size_rf_line.py field solver
D-2	Bias	12 V through 3 x 220R 2512 (73.3 ohm) + 22 nH choke; 104 mA	AOE-3330;WILSON-394	design/rules.tsv R-01..R-03
D-3	Stackup	4 layers: F.Cu RF over In1 GND, In2 +3V3 plane, B.Cu GND pour	BOGATIN-2147	tools/make_template.py
D-4	Routing	RF, clock output, fences, stitching and plane fanouts pre-routed and locked; Freerouting for the rest	ARCH-065;WILLIAMS-2038	tools/make_routes.py
D-5	Edge launch	SMA pads pulled back 0.3 mm; board floor 0.25 mm, 0.5 mm rule for everything else	-	pcb/rffe.kicad_dru
D-6	RF rule frequency	fence/stitching judged at 5 GHz (2nd harmonic of the 2.5 GHz band edge)	ARCH-065;WILLIAMS-2038	design/rf.tsv
D-7	Track floor	board minimum track 0.15 mm (fab capability); Freerouting necks the bias line to 0.15 mm at pins	BROOKS-046	layout check: BIAS IPC-2152 rise on the narrowest segment
D-8	SMA launch	In1 and In2 cut out under the J1/J2 centre pads (1.0 mm beyond each side): the 1.5 mm pad references B.Cu, 47.8 ohm field-solved	JOHNSON03-1307;JOHNSON03-1278	johnson2003; openEMS rf_in; tools/make_template.py
D-9	DC blocks and RF bypass	C1/C2/C3 10 pF C0G 0402: series-resonant near the band with the body ESL, $ X \leq 3.1$ ohm from 2.3 to 2.5 GHz	BOGATIN-167;BOGATIN-2128;ARCH-067;AOE-3330	design/rules.tsv R-05/R-06/R-11

TABLE III
EXECUTION INSTANCES

check	status	method	tool	created
AUTO-CONNECTIVITY	pass	inspection	10.0.5	2026-10-01T05:36:14.688618+00:00
AUTO-DRC	pass	inspection	10.0.5	2026-10-01T05:36:12.040523+00:00
AUTO-EMC	pass	analysis	python 3.13.7; kicad-cli 10.0.5	2026-10-01T05:37:44.894011+00:00
AUTO-ERC	pass	inspection	10.0.5	2026-10-01T05:36:09.915209+00:00
AUTO-LAYOUT	pass	analysis	python 3.13.7; kicad-cli 10.0.5	2026-10-01T05:36:45.496021+00:00
AUTO-PDN	pass	analysis	python 3.13.7; kicad-cli 10.0.5	2026-10-01T05:37:29.648165+00:00
AUTO-RULES	pass	analysis	3.13.7	2026-10-01T05:36:14.969774+00:00
AUTO-SI	pass	simulation	python 3.13.7; kicad-cli 10.0.5; ngspice-47 : Circuit level simulation program	2026-10-01T05:37:34.090480+00:00
AUTO-SIM	pass	simulation	ngspice-47 : Circuit level simulation program	2026-10-01T05:36:13.721656+00:00
AUTO-THERMAL	pass	analysis	python 3.13.7; kicad-cli 10.0.5	2026-10-01T05:37:38.303300+00:00

- University Press, 2015.
- [6] H. Johnson and M. Graham, *High-Speed Signal Propagation: Advanced Black Magic*. Prentice Hall, 2003.
- [7] C. R. Paul, R. C. Scully, and M. A. Steffka, *Introduction to Electromagnetic Compatibility*, 3rd ed. Wiley, 2022.
- [8] T. Williams, *EMC for Product Designers*, 5th ed. Newnes/Elsevier, 2016.
- [9] P. Wilson, *The Circuit Designer's Companion*, 3rd ed. Newnes/Elsevier, 2011.

TABLE IV
CLOSEST CALL OF EACH CHECK, RELATIVE TO ITS LIMIT (EVERY ROW: AUDIT.MD)

check	id	corner	value	units	margin	status
AUTO-EMC	CLK_OUT:excess_over_cispr32_b		-32.2248	dB	26.2248313	PASS
AUTO-LAYOUT	/RF_B:z0@F.Cu/0.32mm		51.4157	ohm	3.5842539	PASS
AUTO-PDN	+3V3:z_pdn_max		0.803463	ohm	0.846537216	PASS
AUTO-RULES	R-01		103.641	mA	6.358925	PASS
AUTO-SI	/CLK_OUT@J3.1:ringback		9.75582	%	10.2441787	PASS
AUTO-SIM	S-2	{ "ctol": "0.8", "rd": "20", "rtol": "0.95" }	-3.09513e+01	dB	20.9513	PASS
AUTO-THERMAL	U2:tj		129.031	C	0.968688	PASS

TABLE V
ITERATION LEDGER

n	change	metric	value	result
1	first wired schematic	netlist_match	MISMATCH (1 label fallback)	discard
2	regenerate after generator fix + fp-lib-table	erc_violations	0	keep
3	first place + Freerouting	drc_violations	137->20	discard
4	plane fanouts + stitching grid + island removal	drc_violations	15	keep
5	SMA pullback 0.3 mm + scoped edge rule	drc_violations	0	keep
6	denser fence/stitching + launch vias + pre-routed clock	layout_pass	53/58->58/58	keep
7	bias 2x150R -> 3x220R	rules_pass	9/10->10/10	keep
8	R1 moved away from U2; 2x3 thermal vias under the GALI tab	tj_U2	130.3->129.0 C	keep
9	SI deck: merge line fragments shorter than the time step	si_pass	stall->4/4	keep
10	openEMS: capacitors as series ESR-ESL-C, only the analysed and ground nets modelled, band-limited pulse, S-settling row	em_energy_db	-11 (never decayed)->-50 in 28.8k steps	keep
11	In1/In2 cut-out under the SMA centre pads	rf_in_S11_db	-6.8->-15.9	keep
12	DC blocks and RF bypass 100 pF -> 10 pF C0G (series-resonant near 2.4 GHz)	R-05_ohm	8.8->3.1	keep
13	EM launch port driven across the coplanar gaps instead of a 1.5 mm tall vertical sheet to B.Cu	rf_in_S11_db	-14.95->-17.1	keep

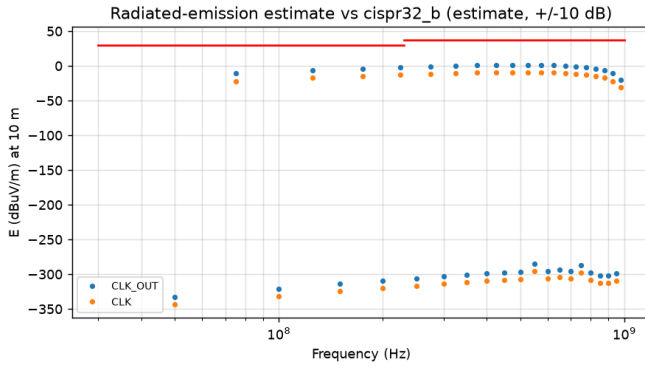


Fig. 4. Radiated-emission estimate against the limit line: emc-cispr32_b

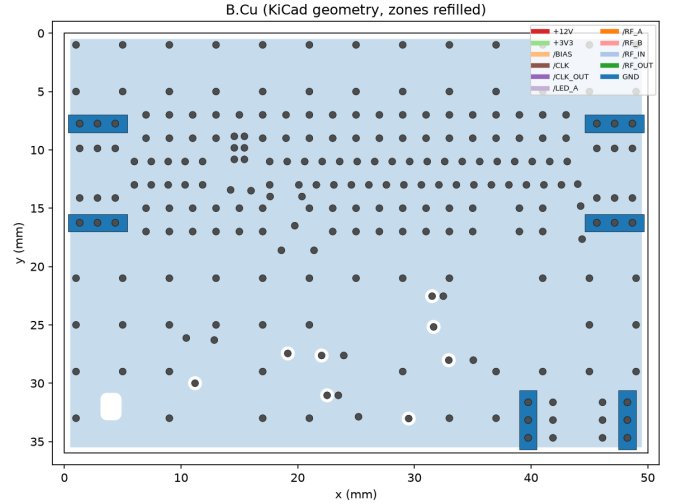


Fig. 5. Copper: copper-B_Cu

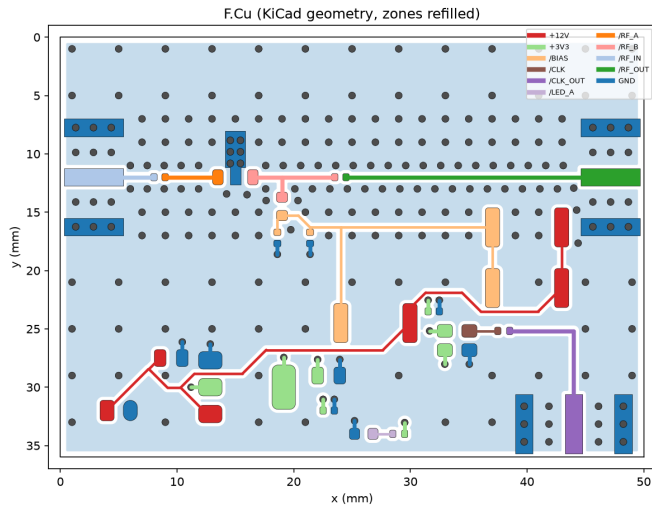


Fig. 6. Copper: copper-F_Cu

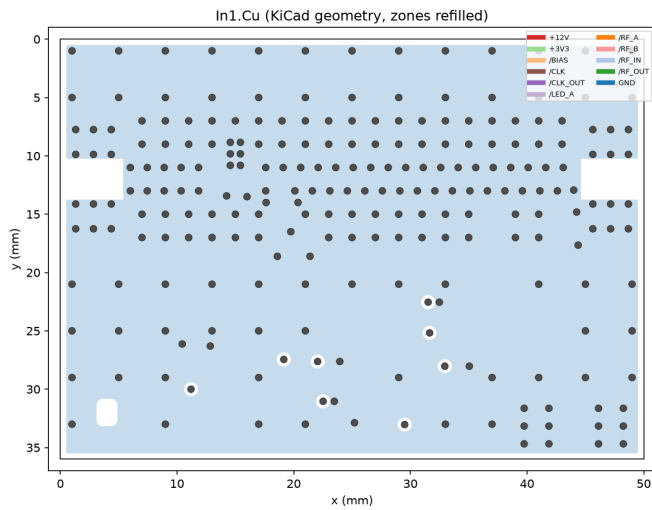


Fig. 7. Copper: copper-In1_Cu

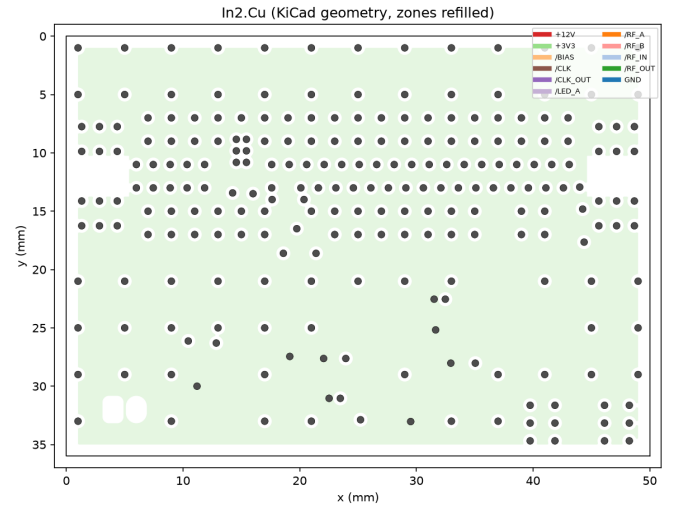


Fig. 8. Copper: copper-In2_Cu

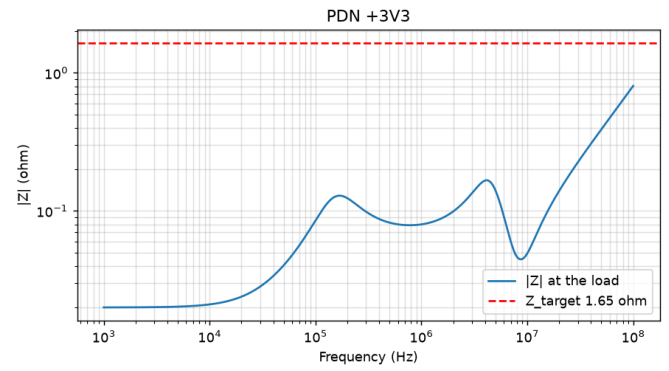


Fig. 9. PDN impedance against Z_{target} : pdn-_3V3

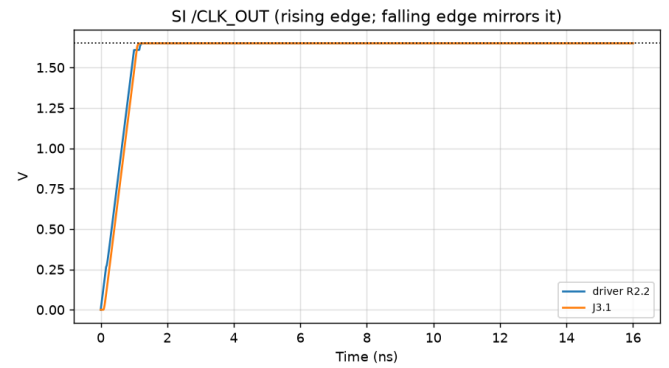


Fig. 10. Signal-integrity waveform: si-CLK_OUT

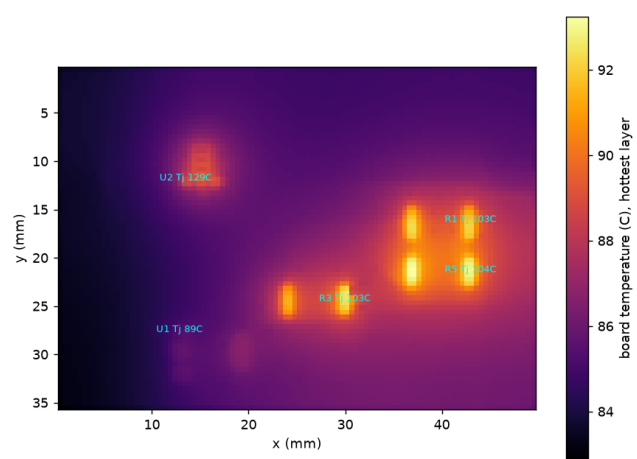


Fig. 11. Board temperature: thermal